Summary of Results

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Summary of PC1 and PC2 Results

PC1

PC1 Volatility

The first analysis used an AR(1) model for the log of annual PC1 volatility.

The original innovations were not IID Gaussian. The normality tests rejected normality, and there was also some dependence at lag 5.

A Yeo-Johnson transformation was then applied. This fixed the normality problem, but some dependence at lag 5 remained. Therefore, the transformation improved the results, but the innovations still did not fully satisfy the IID Gaussian assumption.

PC1 and S&P Volatility

A VAR(1) model was estimated using PC1 bond volatility and S&P volatility. The model produced two innovations: one from the PC1 bond volatility equation and one from the S&P volatility equation.

The PC1 bond volatility innovation was not IID Gaussian. Normality was rejected, and there was some dependence at lag 5.

The S&P volatility innovation performed better. Normality was not rejected, and there was no significant dependence in either the innovation or its absolute value. Therefore, the S&P volatility innovation was consistent with IID Gaussian behavior.

After applying the Yeo-Johnson transformation, the PC1 bond volatility innovation passed the normality tests, but some short-run dependence at lag 5 remained. Therefore, it still did not fully satisfy the IID Gaussian assumption.

The S&P volatility innovation continued to pass the IID Gaussian tests after the transformation.

7-Year Zero-Coupon Treasury Return

The annual log return of the 7-year zero-coupon Treasury was calculated using the 7-year rate in the current year and the 6-year rate in the following year.

The 6-year and 7-year rates were missing for 1992. These observations were kept as missing rather than connecting non-consecutive years.

7-Year Zero-Coupon Rate

An AR(1) model was then estimated for the annual 7-year zero-coupon rate.

The innovations passed the normality tests, and there was no significant dependence in either the innovations or their absolute values at lags 5 and 10. Therefore, the results were consistent with IID Gaussian innovations.

7-Year Rate Model Using PC1 Volatility

The next model included PC1 volatility and normalized the innovation by PC1 volatility.

The normalized innovations passed the normality tests and showed no significant dependence in the innovations themselves. However, there was strong dependence in the absolute innovations.

Therefore, this model did not produce IID Gaussian innovations.

Log 7-Year Rate Model Using PC1 Volatility

The same model was then estimated using the log of the 7-year rate.

This version gave better results. The normalized innovations passed the normality tests, and there was no significant dependence in either the innovations or their absolute values.

Therefore, this model was consistent with IID Gaussian innovations..

PC2

PC2 Volatility

For annual PC2 volatility, the AR(1) model also showed strong persistence over time.

The original PC2 innovations showed no statistically significant serial dependence in either the innovations themselves or their absolute values. Therefore, the results were consistent with IID-type behavior. However, both the Shapiro-Wilk and Jarque-Bera tests strongly rejected normality.

Thus, the original PC2 innovations were consistent with IID behavior but were not Gaussian.

The PC2 innovations were then normalized by the corresponding annual S&P volatility:

$$W_t = \frac{Z_t}{V_t^{SP}}.$$

After normalization, the results improved substantially. There was no statistically significant serial dependence in either W_t or $|W_t|$. In addition, both the Shapiro-Wilk and Jarque-Bera tests failed to reject normality.

Therefore, after normalization by S&P volatility, the PC2 innovations were consistent with an IID Gaussian process.

Main Results

Analysis	Result
PC1 AR(1)	Not IID Gaussian
PC1 AR(1) after Yeo-Johnson	Improved, but some dependence remains

Analysis	Result
PC1-S&P VAR	PC1 innovation is not IID Gaussian; S&P innovation passes
PC1-S&P VAR after Yeo-Johnson	PC1 improved, but some dependence remains
7-year zero-coupon rate AR(1)	Consistent with IID Gaussian
7-year rate model using PC1 volatility	Not IID Gaussian because of dependence in absolute innovations
Log 7-year rate model using PC1 volatility	Consistent with IID Gaussian
PC2 AR(1)	No significant dependence, but not Gaussian
PC2 divided by S&P volatility	Consistent with IID Gaussian

Overall

For PC1, some of the original models did not give IID Gaussian innovations. The Yeo-Johnson transformation helped with normality, but did not completely remove the dependence in the PC1 volatility models.

For the 7-year Treasury rate, the basic AR(1) model worked well. The model using PC1 volatility with the rate in levels did not work. Using the log of the 7-year rate fixed this problem, and the resulting innovations were consistent with IID Gaussian.

For PC2, the original innovations had no significant dependence but were not normal. Dividing them by S&P volatility fixed the normality problem, and the normalized innovations were consistent with IID Gaussian.